

Transfer Settlement Network

Type 10 digits. Pay anyone. No address.

Identity-first, privacy-preserving payments on Solana — with wallet, TIN, and cross-chain routes.

Wallet → Wallet

Wallet → TIN credit

TIN → TIN

TIN → Wallet exit

+ Solana → Creditcoin

THE PROBLEM

Payments have no identity

Crypto made money programmable — but paying someone still means copying a 40-character wallet address and hoping it is right.

There is no portable identity, no privacy boundary, no liability standard. Every payment leaks who paid, who received, and how much.

Lost money

One wrong character and funds are gone. Address poisoning steals from wallets every year.

Lost privacy

Sender, recipient and balance metadata sit on-chain for anyone to read.

Lost time

No labels, no routing, no safety net — every payment is copy, paste, and pray.

Missed mainstream

Normal users will not send money to a random string of characters.

WHY PEOPLE GIVE UP

Today — a wallet address

7xKXtg2CW87d97TXJSDpbD5jBkheTqA83TZRuJesqAsU

× copy · paste · pray



With TSN — a TIN

081-234-5678

✓ type 10 digits · done

VALUE PROPOSITION

**Pay anyone with a 10-digit TIN —
not a wallet address.**

Like a phone number for your money. Resolve it, authorize it, settle it privately.

TIN — portable identity

GPRU — scoped authorization

TCAP — private balance

HOW IT WORKS

One protocol, four routes

Wallet deposit → Wallet exit

The compatibility flow: pay from any wallet to any wallet. Same intent, verification and settlement boundaries as every other route.

Wallet deposit → TIN credit

Fund a TIN from any wallet. The balance lands as a private TCAP credit — the ledger records a commitment, not an amount.

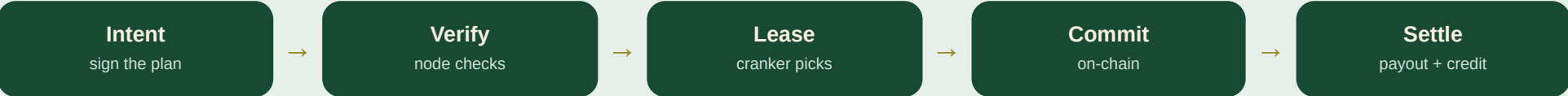
TIN debit → TIN credit

Native identity-to-identity transfer. No wallet address leaves either device; both sides record private credits.

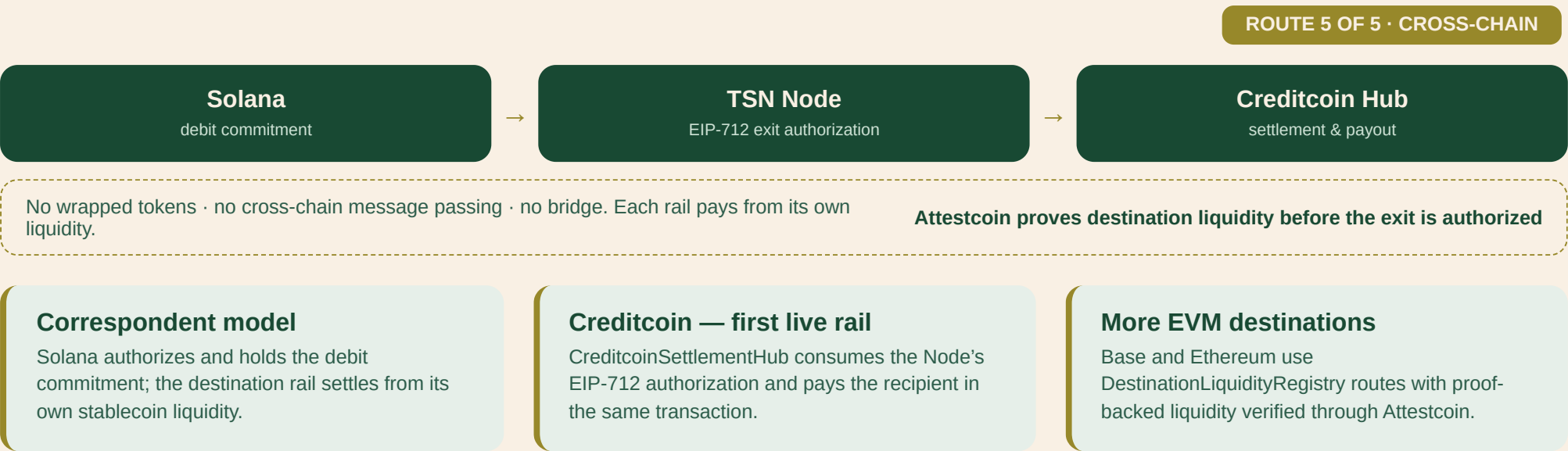
TIN debit → Wallet exit

Cash a TIN debit out to any wallet. A public exit, authorized by a Mother-rooted DNA permit, paid from a cranker vault.

SAME PIPELINE FOR EVERY ROUTE



Beyond Solana — route 5



DEMO

Pay a TIN in three taps

TrustLink Pay — live on Solana Devnet

- Pay with a phone number or 10-digit TIN — not a wallet address
- WhatsApp handles authentication, business verification, and payment notifications
- Private balances via TCAP encrypted snapshots; opaque slot settlement
- Epoch treasury funding; Cranker execution; cryptographic verification

Try it now

trustlink-pay.vercel.app

Docs: trust-link-tsn.mintlify.site · Blog: tsn-protocol.blogspot.com

TRUSTLINK PAY TSN PROTOCOL

Protocol Identity Attestations Settlement Developers

OPERATOR OPEN DAPP

Private, Identity-first, and Confidential Crypto Payments on Solana

TrustLink Pay is built on the Transfer Settlement Network — a decentralized settlement layer for Solana that turns payment intents into private, cryptographically verifiable execution through epoch funding, Cranker execution, and opaque slot settlement. Pay anyone using a phone number or 10-digit TIN instead of a wallet address, with private settlement and optional identity trust context.

[Learn about TSN](#) [Read the repo](#)

IDENTITY LAYER
TIP
Payment identity and optional trust attestations

PRIVACY LAYER
ZK-PRU
Privacy-based protected receiving identities for each Transfer identity

SETTLEMENT
TSN
Payment intents, epochs, Crankers, fees, and receipts

CONFIDENTIAL ASSETS
TCAP
Reserve metadata, commitments, nullifiers, and private coins

Why TrustLink Pay is a Modern Blockchain Solution

A useful blockchain solution should make payments easier for people while preserving the verifiability that makes blockchain settlement valuable. TrustLink Pay does this by combining identity-first payments, stablecoin-native flows, private settlement routing, and developer APIs.

TrustLink Pay is the Web3 payment system users interact with. TSN is the settlement protocol that moves payment work through Crankers. ZK-

Blockchain Payments Without Wallet Addresses
Users can pay a phone number or TIN while the system resolves identity and settlement routes behind the scenes.

Identity-First Blockchain Solution
The Transfer Identity Protocol connects payment identity, verification context, and ZK-PRU route commitments.

Private Blockchain Settlement
TSN separates sender funding from recipient payout so payments do not look

Developer-Friendly Payment Protocol
SDKs expose identity resolution, payment

Creditcoin + Attestcoin, core feature

TSN deploys a TSN-owned Attestcoin Smart Contract on Creditcoin. The contract verifies proof-backed liquidity events from an approved destination EVM route through Creditcoin's native Block Prover precompile.

ATTESTCOIN PROOF PATH

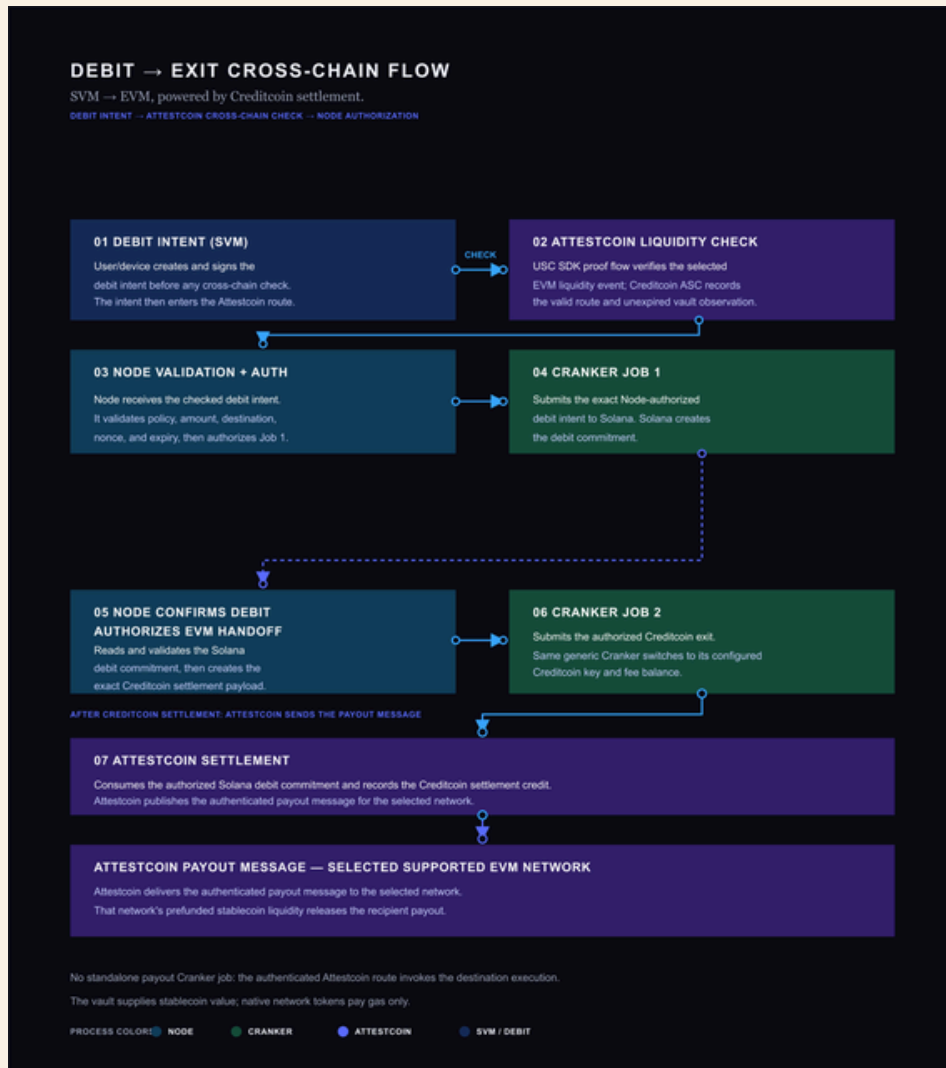


The ASC checks before recording a verified observation:

- Registered source chain & liquidity emitter
- Route & token match the destination payout vault
- Amount within available stablecoin liquidity
- Nonce not consumed (replay protection) & not expired
- LiquidityAvailable event: route · token · amount · nonce · expiry

ARCHITECTURE

Five layers, one payment path



1

Identity

TIN · TIP · GPRU
Portable numbers and scoped routing commitments

2

Intent

Signed payment intents
Off-chain intents binding recipient, amount, mint, nonce, window

3

Authorization

Mother Escrow · DNA
Root authority, permit signing, epoch control

4

Transport

Node · Receiver · Cranker
Verification, durable ingress, exact submission

5

Settlement

Epoch Treasury · TCAP
Liability accounting and confidential balance transitions

The chain sees only commitments — route metadata, private roots, and balances stay off-chain.

Who pays this way

Walletless payments

1.4B+ adults are unbanked yet own a mobile phone (World Bank Findex). A 10-digit identity removes the wallet barrier.

Money over chat

Billions already send money through WhatsApp. TrustLink Pay makes crypto payments feel exactly like that.

Stablecoin settlement

Stablecoin volume keeps compounding on Solana. TSN turns that volume into private, identity-bound settlement.

Demand exists — demand is blocked. People want to pay by phone number; wallets and addresses block them.

BUSINESS

How the network earns

Per-settlement fee

Every TSN payment carries a small fee paid to the fee recipient in the same transaction — already in the settlement path.

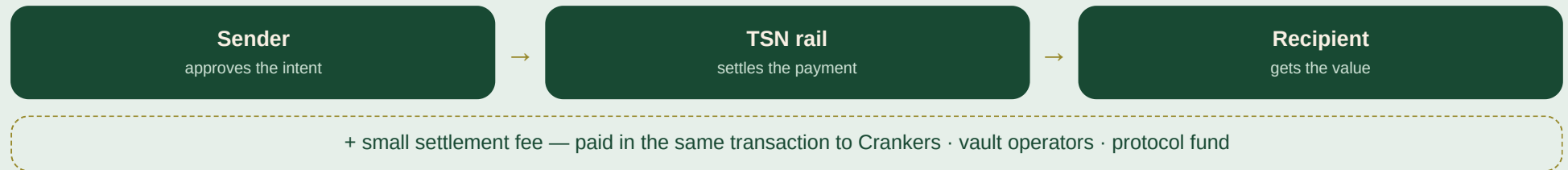
Operator economics

Crankers earn per-submission fees for liquidity and execution; vault operators earn for participation.

Institutional rails

Banks, PSPs and stablecoin issuers plug in through CEIP — licensed settlement for partners.

WHERE REVENUE COMES FROM



TRACTION

Shipped, not promised

Live on Devnet

TIN registrar, GPRU authorization, and TCAP credit transitions ship on Solana Devnet today.

Cross-chain path implemented

Creditcoin-first settlement path with TSN-owned ASC and proof workers — built for this hackathon.

Open monorepo

Contracts, node services, cranker SDK, RPC gateway, and docs — github.com/Trustlink-Labs/TSN-Protocol.

TIN registry · live

GPRU authorization · live

TCAP credit · live

Creditcoin path · built

GROWTH

How we scale

Solana ecosystems

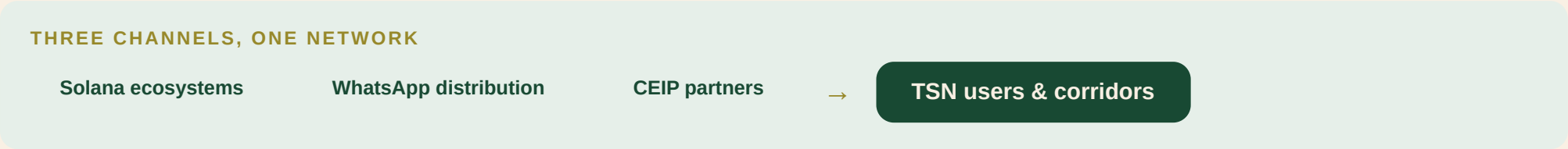
Colosseum, BUIDL, and grant programs put TSN in front of builders who distribute the SDK and run nodes.

WhatsApp distribution

Authentication, business verification, and notifications ride WhatsApp — the UI billions already use.

CEIP partner network

Banks, PSPs and stablecoin issuers integrate settlement through one protocol interface.



What ships next

Now

- Mainnet funding & liquidity bootstrap on Creditcoin
- Additional supported EVM settlement rails
 - Base payout vault drafted

Next

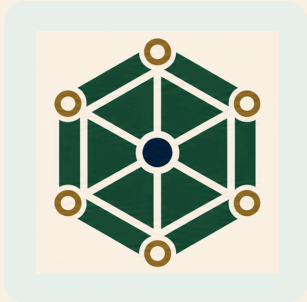
- TCAP confidential debit & exit enablement
- Independent cryptographic verification of Solana commitments

Later

- Wider Attestcoin message-route payout matrix
- Institutional controls & liability tooling
- Partner network integrations via CEIP

TEAM

The team that shipped it



We shipped identity-first settlement — live on Devnet.

TrustLink Labs is an AI-native blockchain lab building open financial infrastructure: privacy, identity, and payment protocols.

- Open monorepo with SDKs, node services, cross-chain contracts & docs
- Owner: Big Dreams Web3 — 1-member team, no blockers, full autonomy
- TrustLink Pay is the reference app proving the routes

Links

github.com/Trustlink-Labs/TSN-Protocol · github.com/bigdreamsweb3/trustlink-pay

trustlink-pay.vercel.app · [@Trustlinklabs](https://twitter.com/Trustlinklabs) · tsn-protocol.blogspot.com · docs: trust-link-tsn.mintlify.site



Payments that settle like institutions — **without the institutions.**

Type a 10-digit TIN. Pay anyone. No address.

Try it — trustlink-pay.vercel.app

Star — github.com/Trustlink-Labs/TSN-Protocol

Read — trust-link-tsn.mintlify.site